From Worksheet to Thinking Task

Redesigning Algebra 2 modeling activities with ClassVoice and the Engagement Blueprint

Cell Phone

Postal Rates

Thermostat



BEFORE: Three scaffolded worksheets

Cell Phone Rates

Compute 3 bills, sketch a graph from 0 to 600 minutes, then write a piecewise rule with domain and range.

Postal Rates

Fill in a rate table, answer cost questions, sketch 0 to 7 ounces, then write the rule.

Setting the Thermostat

Sketch on a pre-drawn grid, find the desired temp, write a function, find domain and range, solve for two differences.

One situation at a time • Step-by-step questions • Pre-made grids and tables • Groups of three

The original worksheets (1 of 2)

Algebra 2
Cell Phone Rates

Name:
Date:
Hour:

You sign a contract for a cell phone. Your contract states that you pay $40.00 per month for the first 400 minutes per month and then you pay $0.15 per minutes for all minutes after 400.

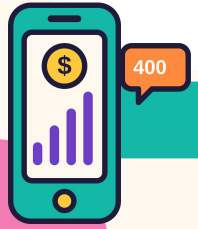
1) How much would your monthly bill be if you talk for 353 minutes?

2) How much would your monthly bill be if you talk for 428 minutes?

3) How much would your monthly bill be if you talk for 0 minutes?

4) Sketch a graph of the contract for minutes ranging from 0 to 600. Use an appropriate scale on your graph.

5) Write a piecewise rule for this situation and state the domain and range for this function.



Cell Phone Rates

Algebra 2
Postal Rates

Name _____

Hour _____

These were the rates for the United States Postal Service for first-class mail.

First-Class Mail Rates
First ounce $0.82
Each additional ounce: $0.29

1.Complete the table below.

First-Class Mail Rates

Weight not over (ounces)	Rate
1	
2	
3	
4	
5	
6	

2.How much would you pay for a first-class mailing if it weighed 2.6 ounces?

3.How much would you pay for first-class mailing if it weighed 2 ounces?

4.How much would you pay for first-class mailing if it weighed 2 ounces?



Postal Rates, page 1

Algebra 2
Setting the Thermostat

Name _____

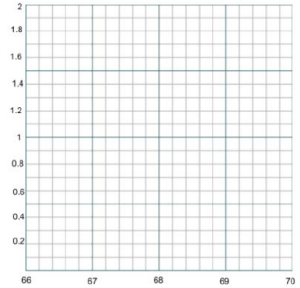
Hour _____

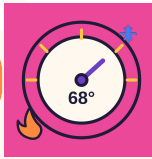
A thermostat is set at T° (Desired Temperature) but it may vary (Actual Temperature).

Actual Temperature	Difference between Desired Temperature and Actual Temperature
66.	2
66.4	1.6
66.8	1.2
67.2	0.8

Actual Temperature	Difference between Desired Temperature and Actual Temperature
68.8	0.8
69.2	1.2
69.6	1.6
70.	2.

a. Sketch a graph of this situation.

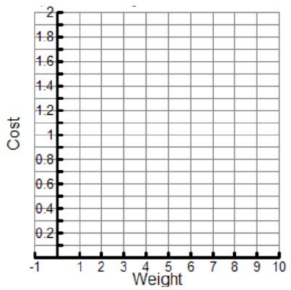




Thermostat, page 1

The original worksheets (2 of 2)

5. Sketch a graph of the postal mailing rates for first class from 0 to 7 ounces.



6. Write a rule for this situation and state the domain and range for this function.



Postal Rates, page 2

b. If the pattern in the data continues, what is the desired temperature?

c. Write a function that would model this situation.

d. State the domain and range of the function given this situation. Explain your reasoning.

e. If the difference between the desired temperature and the actual temperature is 0.23, what is the actual temperature of the room? Justify your solution.

f. If the difference between the desired temperature and the actual temperature is 2.14, what is the actual temperature of the room? Justify your solution.

Thermostat, page 2



ClassVoice AMT Prediction

Acquisition: HIGH

Students calculate costs and values, fill a table, sketch specified graphs, and write rules with domain and range. The worksheet directs each step.

Meaning Making: LOW

The materials decide what to compute, which intervals to graph, and when to write the rule. Students do not decide what to notice or how to structure evidence.

Transfer: ABSENT

No genuinely new situation with limited prompting. Each context is a guided, first-pass task.

Cognitive ownership sits with the materials, capping demand at procedures without connections.

Positioning Blueprint: 1 of 7 in place

1a Content and concept	**Partial**	Content is clear; breakpoints and step vs continuous are not made explicit.
1b Multiple artifacts and patterns	**Absent**	Three contexts processed separately; no comparison.
1c High-cognitive prompt	**Absent**	Numbered sub-questions prescribe the sequence.
2a Participation structure	**Present**	Teacher plans groups of three.
2b Large blank space	**Absent**	Handouts are segmented into lines and blanks.
2c Extended small-group discussion	**Absent**	Group work focused on completing steps.
2d Whole-class reification	**Absent**	No plan to synthesize across the three tasks.

ClassVoice suggestion: smallest change, biggest impact

1. Remove the step-by-step questions
2. Give only the three situation descriptions
3. Add one high-level prompt on blank whiteboards or paper
4. Constrain output: one page per context plus a brief comparison note

Whole-class synthesis: name students' noticed features first (breakpoint, flat-then-slope, steps, distance from a target), then attach formal terms (piecewise, step function, absolute value, domain, range).

AFTER: One task, three situations

Represent each situation so a peer can quickly see the underlying rule. Choose representations that make clear (i) where the behavior changes or the target lies, and (ii) how the output behaves before, at, and after that point. Include a graph and a written rule in your own words, then formalize with symbols. Finally, compare the three: What is the same and what is different about their structures? Justify your choices.

Cell phone plan

Postal rates

Thermostat data

Work on blank whiteboards or blank paper

The New Prompts / Task

Algebra 2 — Modeling Situations

Situation 1: Cell Phone Plan

You sign a contract for a cell phone. Your contract states that you **pay $40.00 per month for the first 400 minutes per month**, and then you **pay $0.15 per minute for all minutes after 400.**

Situation 2: Postal Rates

These were the rates for the United States Postal Service for first-class mail:

First-Class Mail Rates	
First ounce	$0.82
Each additional ounce	$0.29

Situation 3: Setting the Thermostat

A thermostat is set at T° (Desired Temperature), but the Actual Temperature may vary. The table below shows the difference between the desired temperature and the actual temperature:

Actual Temp.	Difference	Actual Temp.	Difference
66.	2	68.8	0.8
66.4	1.6	69.2	1.2
66.8	1.2	69.6	1.6
67.2	0.8	70.	2.

Task

Represent each situation so a peer can quickly see the underlying rule.

Choose representations that clearly identify

(i) the critical value of the input (the point of discontinuity, threshold, or target value)

(ii) the behavior of the function on each side of that value

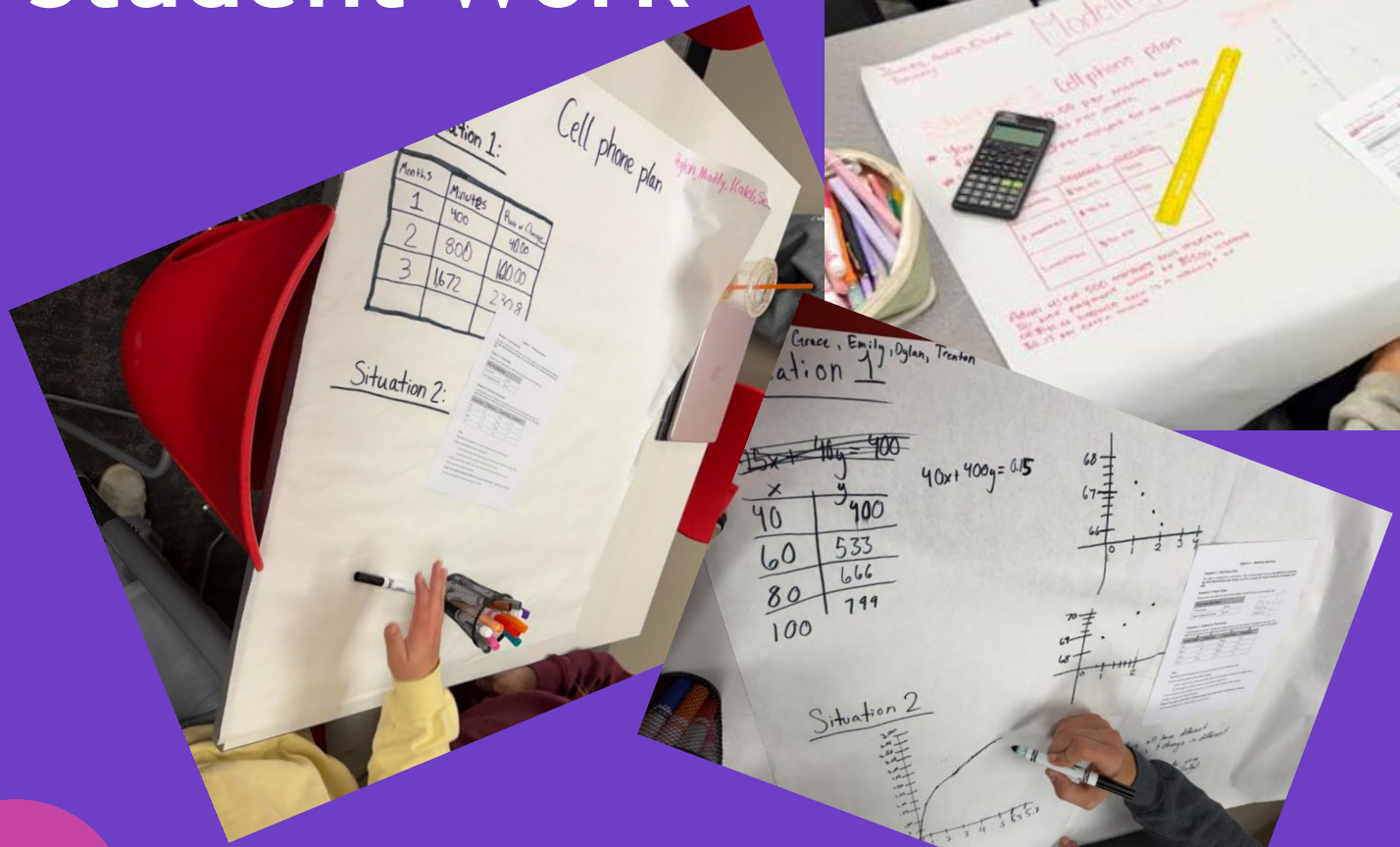
Include a graph and a written rule in your own words, then formalize with symbols.

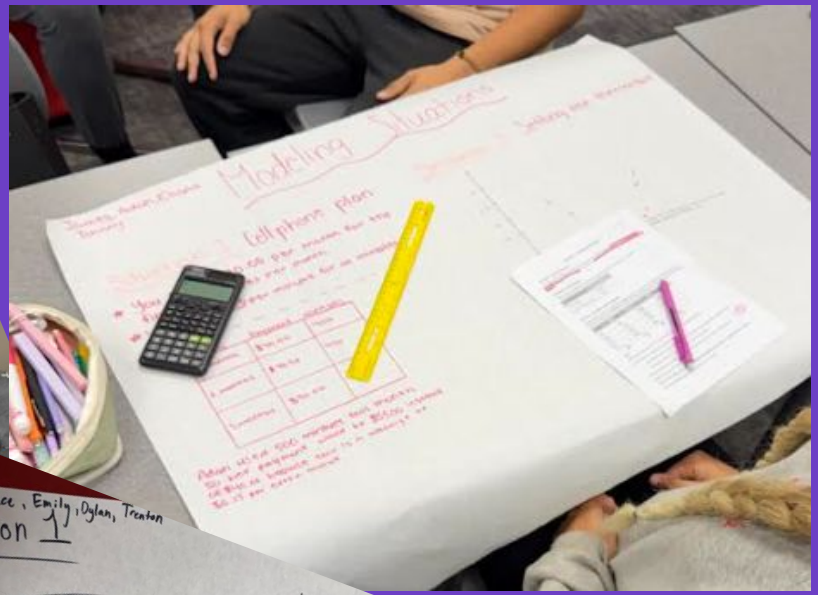
Finally, compare the three situations:

What is the same and what is different about their structures? Justify your choices.

Use blank whiteboards or blank paper for your work.

Student Work





The Blueprint: Before → After

1a Key Concept

Before: breakpoints and step vs continuous never named.
After: students surface them; class attaches formal terms.

1b Compare Patterns

Before: three contexts handled separately.
After: asked to compare all three structures.

1c Open Prompt

Before: numbered sub-questions set the sequence.
After: one open prompt; students choose representations.

2a Grouping

Before: groups of three already planned.
After: kept; groups build a shared rule for a peer.

2b Blank Space

Before: lines and blanks.
After: blank whiteboards or paper.

2c Discussion

Before: completing steps.
After: sustained comparing and justifying.

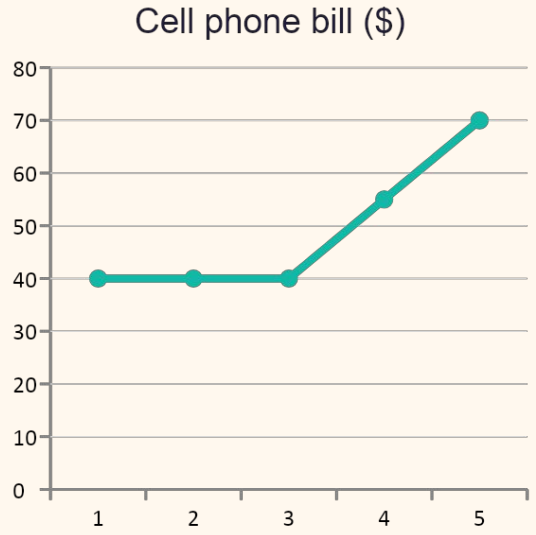
2d Class Reify

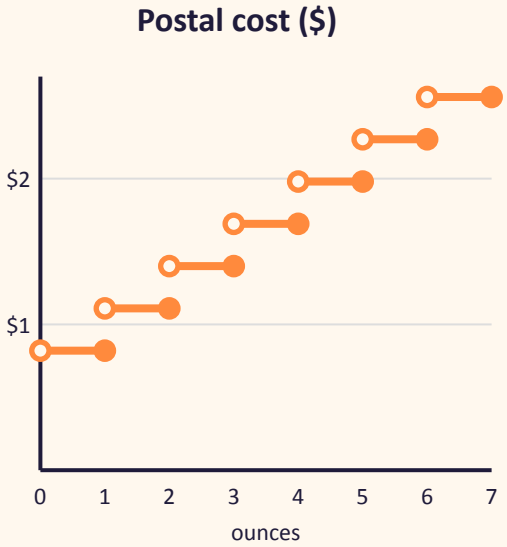
Before: no synthesis plan.
After: name noticed features first, then formal terms.

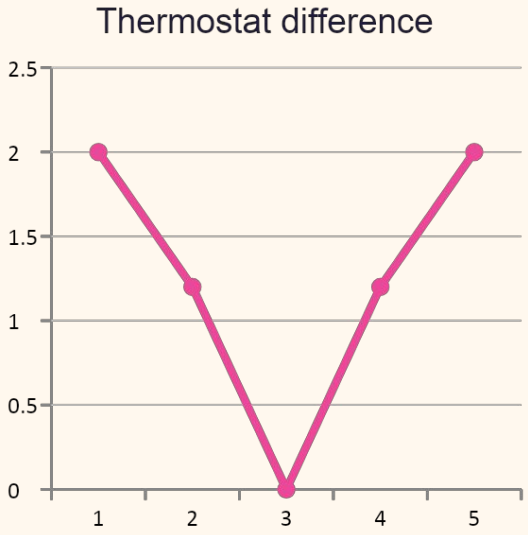
Students Take Notes

Before: the worksheet held the answers.
After: students take notes on the class synthesis.

The three situations at a glance







Flat, then rises at 400 minutes

Jumps at each new ounce

V-shape: lowest at 68

Same or different?

SAME	DIFFERENT
A key point where the rule changes or the target lies Different rules before and after it Piecewise-style thinking and symbols	Phone: continuous, slope changes at 400 Postal: step function with jumps Thermostat: symmetric V around the target

The whole-class reify makes this comparison visible

Try it in your classroom

1. Ask a question with no single procedure
2. Group on purpose
3. Hand out blank space, not a grid
4. Leave time to talk
5. Reify together, then notes